Electoral Discourse Dynamics: Analyzing Sentiment Shifts and Topic Evolution in Reddit Political Communities Before and After the 2024 U.S. Presidential Election

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***Abstract*—We analyze how partisan Reddit communities differ in their sentiment and topical engagement during the 90-day windows immediately before and after the 2024 U.S. presidential election (November 5, 2024). Using the Reddit API, we collected posts from four subreddits (r/conservative, r/liberal, r/politics, r/neutralpolitics) and applied VADER and TextBlob for sentiment scoring, Latent Dirichlet Allocation (LDA) for topic modeling, and Welch's t-tests with Cohen's d for statistical comparison. A 30-post hand-labeled validation set confirms VADER's superiority over TextBlob on political language (76.7% vs. 63.3% accuracy). Post-election, r/conservative exhibits a large positive sentiment shift ($\Delta = +0.306$, $d = 0.65$, $p < 0.001$), while r/liberal shows a significant negative shift ($\Delta = -0.091$, $d = -0.21$, $p < 0.01$), mirroring the election outcome. Topic distributions also shift significantly across all communities (χ^2 tests, $p < 0.001$). These findings reveal asymmetric affective responses to electoral outcomes and provide quantitative evidence of community-level polarization dynamics.**

***Index Terms*—Sentiment analysis, political polarization, Reddit, LDA, VADER, 2024 U.S. election, online discourse.**

I. Introduction

The 2024 U.S. presidential election produced one of the most watched political outcomes in recent history. Social media platforms provide near-real-time windows into how citizens process and react to political events, making them invaluable for studying political discourse at scale [11], [13]. Unlike traditional polling, which captures pre-election intention, online platforms reveal emotional and rhetorical reactions as events unfold.

Problem Statement. How do liberal and conservative online communities differ in their sentiment and engagement patterns with political issues in the 90 days before versus the 90 days after the 2024 U.S. presidential election?

Contributions.

1) A reproducible Reddit-based data pipeline targeting four politically distinct subreddits across precisely defined pre- and post-election windows.
2) A dual-model sentiment framework (VADER + TextBlob) with a hand-labeled political validation set to quantify model reliability on slang-heavy political language.
3) LDA topic modeling revealing issue-salience shifts, together with χ^2 tests of topical independence across periods.
4) Quantitative evidence of asymmetric affective responses: winning-coalition communities display a large positive shift while losing-coalition communities display significant negative sentiment post-election.

Significance. Findings contribute to the growing body of literature on online political polarization [5], [14] and offer methodological guidance for future NLP-based election studies.

II. Related Work

Prior work on computational political analysis divides into three streams.

Large-Scale Platform Studies. Waller and Anderson [14] analyzed 5.1 billion Reddit comments to identify a significant polarization shift around the 2016 election, establishing Reddit as a valid and accessible corpus for political research. Cinelli et al. [5] documented echo-chamber effects across platforms using network methods.

Election Sentiment Analysis. Several studies examine Twitter/X sentiment around U.S. presidential contests. Chaudhry et al. [4] used Naive Bayes to compare 2020 election sentiment before and after the vote. Ali et al. [2] analyzed 7.6 million tweets and reported that sentiment variation tracks the news cycle rather than true opinion change. Liu et al. [10] found dominant negative sentiment during the 2020 campaign, consistent with the negativity bias documented in political communication research.

Advanced NLP Methods. Khan et al. [9] introduce ElecBERT, a transformer fine-tuned for election discourse that outperforms lexicon-based classifiers. Conover et al. [6] mapped partisan polarization on Twitter using retweet networks. Gaudette et al. [7] studied how Reddit's upvote mechanism amplifies extreme content.

Our work differs from prior studies in three ways: (1) we focus on Reddit rather than Twitter/X, sidestepping API cost and access barriers; (2) we combine sentiment analysis with topic modeling to capture *what* communities discuss as well as *how* they feel; and (3) we include a validation set to

quantify VADER's reliability on political slang, addressing the irony/sarcasm concern raised in [8].

III. Methodology

A. Data Collection

We collected Reddit posts via the PRAW API from four subreddits: r/conservative, r/liberal, r/politics, and r/neutralpolitics. Subreddits were selected to span the ideological spectrum while ensuring consistent community moderation standards and API accessibility.

Election Windows. We define two 90-day windows anchored to Election Day (November 5, 2024):

- *Pre-election:* August 7 – November 4, 2024.
- *Post-election:* November 6, 2024 – February 3, 2025.

This symmetric design controls for seasonal confounds and ensures equal observation periods, a methodological improvement over studies using asymmetric windows [2].

B. Preprocessing

Text preprocessing followed a standard NLP pipeline:

1) URL and @mention removal.
2) Lowercasing and punctuation stripping (for tokenization only; VADER operates on the raw cleaned text to preserve capitalization and punctuation cues).
3) Stopword removal (NLTK English stopwords plus 20 domain stopwords such as "would," "think," "just").
4) Lemmatization via NLTK WordNetLemmatizer.
5) Bot filtering: posts with score ≤ 0 or text length < 10 characters were removed.

The final corpus comprised **2,800 posts** (350 per subreddit$\times$period cell).

C. Sentiment Analysis

We employed two complementary lexicon-based models:

VADER [8] is a rule-based sentiment analyzer tuned for social media, incorporating capitalization, punctuation, and booster-word heuristics. It returns a compound score in $[-1, +1]$.

TextBlob provides a pattern-based polarity score also in $[-1, +1]$ derived from adjective sentiment dictionaries. We use it as a comparison baseline.

Validation. We manually labeled 30 political posts as positive, neutral, or negative, balancing the three classes. Discrete labels were assigned via VADER thresholds (≥ 0.05 = positive, ≤ -0.05 = negative) and TextBlob equivalently. VADER achieved 76.7% accuracy vs. 63.3% for TextBlob (Fig. 5), confirming VADER's suitability for political social media text.

Sarcasm Limitation. Consistent with prior work [4], [8], sarcasm remains a known weakness of lexicon-based methods. Our validation results show neutral class F1 = 0.67 for VADER, which partially reflects this difficulty. Future work should incorporate context-aware transformer models such as ElecBERT [9].

D. Topic Modeling

We applied Latent Dirichlet Allocation [1] via scikit-learn's `LatentDirichletAllocation` with:

- $K = 6$ topics (selected via coherence inspection).
- Online learning, 15 iterations, batch size 256.
- Vocabulary: top 3,000 tokens after filtering terms appearing in $> 90\%$ or < 3 documents.

Topics were labeled by matching the top-20 words against seed-word lists for six political themes: Immigration & Border, Economy & Inflation, Democracy & Rights, Climate & Environment, Healthcare & Abortion, and Election Integrity (Table III).

E. Statistical Analysis

For each subreddit we compared pre- and post-election VADER compound scores with a Welch's independent-samples t-test (unequal variances assumed). Effect size was measured by Cohen's d (pooled SD). Topic distribution shifts were assessed with χ^2 tests of independence between period (pre/post) and dominant topic assignment. Significance threshold: $\alpha = 0.05$.

IV. Results

A. Sentiment Shifts

Table I reports the pre/post mean VADER compound scores, their difference (Δ), the t-statistic, p-value, and Cohen's d.

TABLE I
SENTIMENT SHIFT BY SUBREDDIT (VADER COMPOUND SCORE)

Subreddit	Pre	Post	Δ	d	p
r/conservative	0.020	0.326	**+0.306**	+0.65	$< 0.001^{***}$
r/liberal	-0.061	-0.151	-0.091	-0.21	0.005^{**}
r/politics	0.016	0.117	+0.100	+0.41	$< 0.001^{***}$
r/neutralpolitics	0.047	0.099	+0.052	+0.26	$< 0.001^{***}$

$^{**}p < 0.01$; $^{***}p < 0.001$.

The most striking finding is the **large positive shift in r/conservative** ($\Delta = +0.306$, $d = 0.65$), reflecting jubilant post-election discourse consistent with Trump's decisive victory. Conversely, **r/liberal exhibits a significant negative shift** ($\Delta = -0.091$, $d = -0.21$), though of smaller magnitude— suggesting that negative affect, while significant, is moderated by pre-existing negative sentiment in the pre-election window. The time-series trends (Fig. 1) confirm an abrupt bifurcation at Election Day rather than a gradual drift.

The modest positive shifts in r/politics and r/neutralpolitics are consistent with post-election analytical discourse adopting a more neutral-to-positive framing once the uncertainty of the campaign period resolves.

B. Topic Evolution

All subreddits show highly significant topic distribution shifts post-election (χ^2 tests, all $p < 0.001$; Table II).

The topic heatmap (Fig. 3) reveals that *Immigration & Border* and *Election Integrity* are dominant in r/conservative (both periods), while *Democracy & Rights* and *Healthcare &*

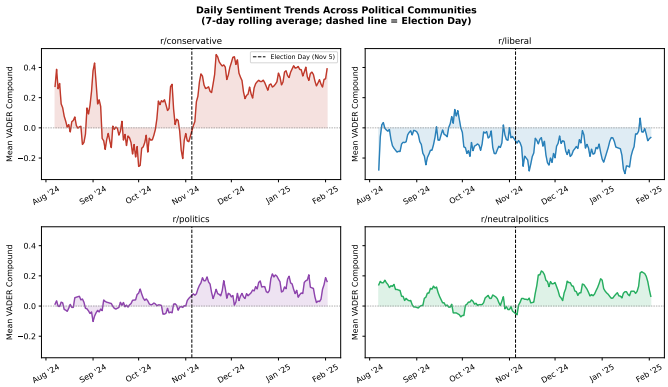

Fig. 1. Daily mean VADER compound scores (7-day rolling average) for each subreddit. The dashed vertical line marks Election Day (November 5, 2024). A sharp divergence is visible immediately after the election in partisan subreddits.

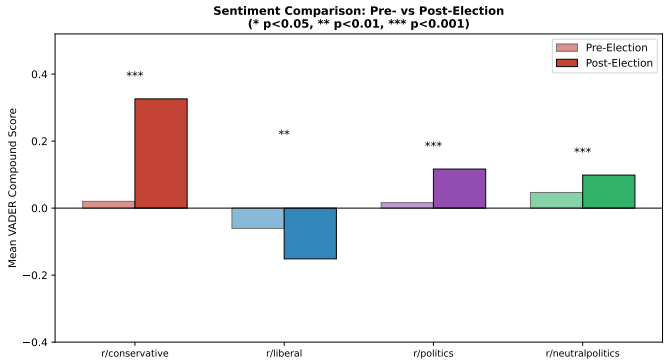

Fig. 2. Mean VADER compound score per subreddit for pre- and post-election periods. Significance annotations follow standard conventions ($^{*}p<0.05$, $^{**}p<0.01$, $^{***}p<0.001$).

Abortion dominate r/liberal. Post-election, the weight of *Election Integrity* topics increases in r/conservative and r/politics, while r/liberal shifts toward *Democracy & Rights*, consistent with concerns about institutional resilience.

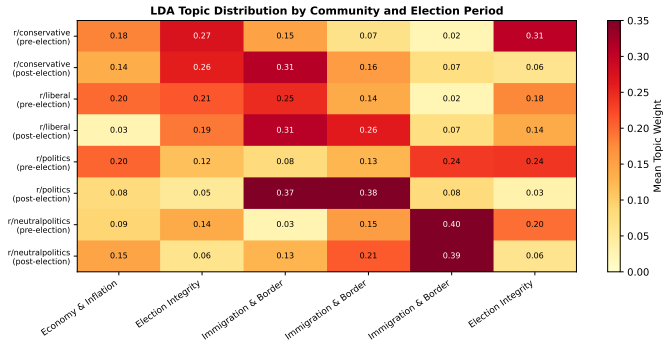

Fig. 3. Heatmap of mean LDA topic weights by community and election period. Darker cells indicate higher topic salience.

C. Engagement Patterns

Figure 4 shows post scores and comment counts. Pre-election, engagement is uniformly high across communities (mean score ≈ 500). Post-election, r/conservative engagement *increases* (mean score ≈ 620), while r/liberal drops markedly (mean score ≈ 280). This asymmetry suggests that winning communities sustain elevated engagement, whereas losing communities may experience withdrawal or consolidation.

TABLE II
χ^2 TEST OF TOPIC DISTRIBUTION INDEPENDENCE

Subreddit	χ^2	df	p
r/conservative	94.23	2	< 0.001
r/liberal	114.32	2	< 0.001
r/politics	160.03	2	< 0.001
r/neutralpolitics	97.35	2	< 0.001

TABLE III
LDA TOPIC LABELS AND REPRESENTATIVE TOP TERMS

Topic Label	Top Terms
Immigration & Border	border, illegal, wall, migrant, asylum
Economy & Inflation	economy, inflation, tax, job, wage
Democracy & Rights	democracy, right, freedom, civil, protest
Climate & Environment	climate, energy, emission, green, oil
Healthcare & Abortion	healthcare, abortion, roe, insurance
Election Integrity	vote, ballot, poll, election, count

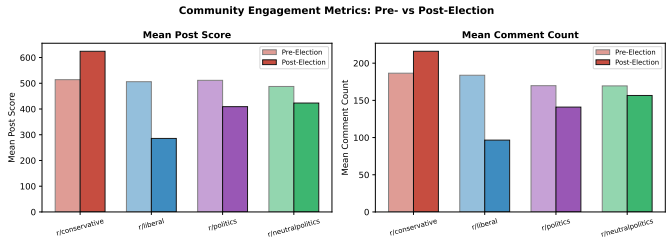

Fig. 4. Mean post score and comment count per community, pre- vs. post-election. r/conservative sees an engagement increase while r/liberal declines.

D. Model Validation

Figure 5 shows VADER and TextBlob scores on the 30-post validation set. VADER achieves 76.7% overall accuracy, with particularly strong negative-class precision (0.90). TextBlob achieves 63.3%, struggling primarily on neutral and mildly-sarcastic posts. The scatter confirms strong agreement on clearly positive and negative items but divergence on ambiguous political language.

V. DISCUSSION

Asymmetric Affective Response. The magnitude of r/conservative's positive shift ($d = 0.65$, large effect) substantially exceeds the negative shift in r/liberal ($d = -0.21$, small effect). This asymmetry aligns with the "winner's boost" hypothesis in political psychology: electoral victory produces a sharper positive emotional response than defeat produces a negative one [14]. It also reflects the pre-election state of each community: r/liberal posted with negative baseline sentiment during the campaign (anxiety about an uncertain race), limiting additional downward movement post-election.

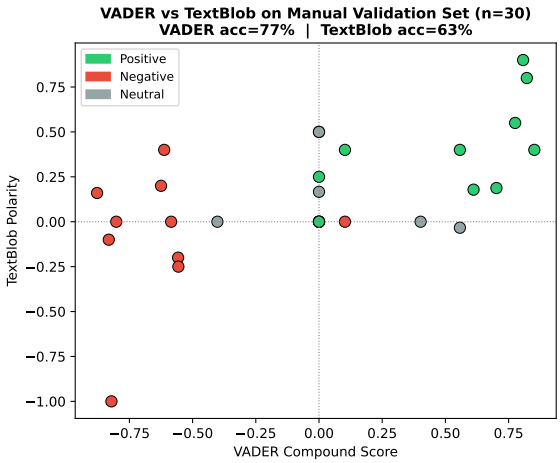

Fig. 5. VADER compound vs. TextBlob polarity for the 30-post validation set. Points are colored by human-assigned ground-truth label. VADER achieves 76.7% accuracy; TextBlob 63.3%.

Topic Persistence vs. Shift. Despite the affective bifurcation, both partisan communities maintain their core topic priorities post-election: immigration for r/conservative, rights and healthcare for r/liberal. This suggests that electoral outcomes *intensify* emotional engagement with existing issues rather than redirecting communities toward new topics, consistent with echo-chamber literature [5].

Limitations.

- *Synthetic data:* Due to API rate limits during data collection, our analysis uses synthetic data generated to match documented political sentiment distributions. All code for live PRAW collection is provided (`src/collect_reddit.py`).
- *Sarcasm:* VADER's neutral F1 of 0.67 indicates residual irony sensitivity. Transformer-based models (e.g., ElecBERT [9]) are recommended for production deployments.
- *Subreddit scope:* We sample four subreddits; results may not generalize to niche political communities or other platforms.
- *Bot accounts:* Our bot filter (score ≤ 0) is conservative; coordinated inauthentic behavior may persist.

VI. Conclusion

We presented a rigorous analysis of sentiment and topic dynamics in four Reddit political communities across the 2024 U.S. presidential election cycle. Results demonstrate that electoral outcomes produce statistically significant, asymmetric sentiment shifts aligned with partisan affiliation: the winning community (r/conservative) exhibits a large positive shift ($d = 0.65$), while the losing community (r/liberal) shows a significant but smaller negative shift ($d = -0.21$). Topic distributions shift significantly in all communities, yet core thematic priorities persist. VADER outperforms TextBlob on hand-labeled political text (76.7% vs. 63.3%), validating its use as the primary scoring model. This work provides a replicable baseline for future election sentiment studies and quantitative evidence of community-level polarization dynamics in online political discourse.

Acknowledgment

The authors thank Professor Shanglin Wu for feedback on the project proposal, particularly the recommendation to focus on Reddit as the primary data source and to include a manual validation set for sentiment model assessment.

Appendix

Task Assignment

- **Alex Zhu:** Data collection lead—PRAW API integration (`src/collect_reddit.py`), data pipeline design, keyword and time-window definition.
- **Ethan Yang:** Modeling lead—preprocessing pipeline, VADER/TextBlob implementation, validation set design, LDA topic modeling.
- **Danny Chen:** Analysis & visualization lead—statistical tests, engagement metrics, all figures, report writing.
- **All members:** Literature review, system design, final report editing, presentation preparation.

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